

Research Article

Molecular characterization of *rpoB* gene mutations in rifampicine-resistant *Mycobacterium tuberculosis* isolates from tuberculosis patients in Belarus

Leonid P. Titov^{1,2}, Saeed Zakerbostanabad^{2,3}, Veranika Slizen^{1,2}, Larisa Surkova⁴, Mohammad Taghikhani^{4,5} and Ahmadreza Bahrmand³

¹Belarusian Research Institute for Epidemiology and Microbiology, Minsk, Belarus

²Belarusian State Medical University, Minsk, Belarus

³Pasteur Institute of Iran, Tehran, Iran

⁴Institute of Pulmonology and Tuberculosis of Minsk, Minsk, Belarus

⁵Biochemistry Department, School of Medical Sciences, Tarbiat Modarres University, Tehran, Iran

The aim of this study was to investigate the frequency, location and type of *rpoB* mutations in *Mycobacterium tuberculosis* isolated from patients in Belarus. Tuberculosis cases are increasing every year in Belarus. Moreover, resistance to anti-tuberculosis drugs, especially to rifampicine, has increased. In this study, 44 rifampicine-resistance *M. tuberculosis* clinical isolates (including multidrug-resistant isolates) were subjected to DNA sequencing analysis of the hypervariable region (hot-spot) of the *rpoB* gene originating from different geographical regions in Belarus. Sixteen different types of mutations were identified. The most common point mutations were in codons 510 (47.7%), 526 (45.5%), 523 (40.86%) and 531 (29.5%). Eleven isolates (27.7%) carried one mutation and 23 (52%), 7 (16%), 3 (7%) of isolates carried 2, 3 and 4 mutations, respectively. A characteristic, prominent finding of this study was high frequency of multiple mutations in different codons of the *rpoB* gene (27.7%) and also the detection of unusual types of mutations in the 510 codon, comprising CAG mutations (deletion or changing, to CTG, CAC or CAT). In our study, the change TTG in codon 531 was found in 92% of isolates and the change TGC in 8% of isolates. A TAC change in codon 526 was not found, but the GAC change was found in all isolates. Isolates of *M. tuberculosis* isolated in Belarus were characterized by the wide spectrum of the important mutations and might belong to the epidemic widespread clones.

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1 Introduction

Mycobacterium tuberculosis is one of the most pervasive, morbid, and lethal human pathogens, which causes yearly about 8 million new cases of tuberculosis (TB) world-

wide, with between 2 to 3 million deaths being reported every year [1–3]. The WHO has estimated that 1 billion people will be newly infected, 200 million will get sick, and 35 million will die from TB in near future.

The frequency and spectrum of antibiotic resistance have increased worldwide during the past few decades. This increase has been attributed to a combination of microbial characteristics, the selective pressure of antimicrobial use, and social, economical changes that enhance the generation and transmission of resistant organisms [4, 5]. In 2004, 32.6% [9.6% multidrug (MDR)+23% anti-TB drug resistance] of new cases and 81.6% (28.6% MDR+53% anti-TB drug resistance) of previously treated

Correspondence: Dr. Saeed Zakerbostanabad, Clinical Microbiology Department, Belarusian Research for Microbiology and Epidemiology, St. Filimonova 23, Minsk, 220114 Belarus
E-mail: saeedzaker20@yahoo.com
Fax: +375 17 26 73 467

Abbreviation: TB, Tuberculosis

TB cases were found to be resistant to anti-*M. tuberculosis* drugs in Belarus. Disease rates reported by the Belarusian Health Ministry showed that 50.6 out of 100 000 people were infected in 2004. Primary drug resistance has been monitored since 1997, with decreasing frequencies occurring up to 2002 [6]. The upsurge of TB has been accompanied by a rise in the frequency of *M. tuberculosis* isolates that are resistant to one or more primary anti-TB drugs (41%) [6]. Early diagnosis of the disease and the rapid identification of resistance to primary anti-TB drugs are essential for the efficient treatment and control of MDR isolates. Resistance to rifampicine is a key factor in determining the effectiveness of the currently recommended standard treatment regimens [7, 8]. The elucidation of the mechanism of action of these drugs, which was accomplished recently, has led to the development of new rapid diagnostic methods [9–11]. Of the rifampicine-resistant isolates of *M. tuberculosis*, 96% possess genetic alterations within the 81-bp fragment of the *rpoB* gene that encodes the β subunit of DNA-dependent RNA polymerase, which is called the rifampicine resistance-determining region (RRDR) [10, 12, 13]. Many different allelic variations have been detected within this region [14], and specific *rpoB* genotypes are known to be associated with high-level rifampicine resistance [1, 3, 8, 9].

2 Materials and methods

2.1 Mycobacterial isolates and growth condition

The 44 rifampicine-resistant isolates isolated from sputum of patients with active pulmonary tuberculosis were obtained from various geographical regions of Belarus (Minsk, Mogilev, Gomel, Grodno, Brest, Vitebsk) from October 2004 to July 2005. All the isolates were cultured on Löwenstein-Jensen solid medium, and were identified to the species level using 2-thiophene carboxylic acid and PN99B (paranitrobenzoic acid)-selective media or by standard biochemical procedures.

2.2 Susceptibility, methods and minimum inhibitory concentrations

All isolates were resistant to at least one primary anti-TB agent. Drug susceptibility was determined using the ab-

solute concentration method (1 μ g/mL isoniazid, 40 μ g/mL rifampicine, 10 μ g/mL streptomycin, or 2 μ g/mL ethambutol) on Löwenstein-Jensen medium with H37Rv strain of *M. tuberculosis* as the positive control, and using the BACTEC system. With the absolute concentration method, resistance was defined as growth on solid media containing graded concentrations of drugs with more than 20 CFU at a specific drug concentration. The break-points for isoniazid were at 1 μ g/mL on Löwenstein-Jensen medium and 0.1 μ g/mL on the BACTEC system; for rifampicine, 40.0 μ g/mL on Löwenstein-Jensen medium and 2.0 μ g/mL on the BACTEC system.

2.3 DNA extraction

For standard PCR identification and *rpoB* gene amplification, the 44 rifampicine-resistant isolates were collected and DNA extraction was performed [8] (TAE 200 + 5% chelex100; 45 min 95°C). DNA isolated from standards isolates of *M. tuberculosis* Academia and *Mycobacterium bovis* BCG were used as controls. A 411-bp fragment of the *rpoB* gene was amplified by PCR with the following synthetic oligonucleotide primers: *rpoB*-F (5'-TACG-GTCGGCGAGCTGATCC-3') and *rpoB*-R (5'-TACG-GCGTTTCGATGAACC-3'). The following thermocycler parameters were used: initial denaturation at 94°C for 5 min; 42 cycles of denaturation at 94°C for 1 min; primer annealing at 57°C for 1 min; extension at 72°C for 1 min; and a final extension at 72°C for 10 min. The product was checked and purified by gel electrophoresis (Fig. 1) and the purified *rpoB* fragment was amplified. The resultant DNA was used for direct sequencing.

2.4 DNA sequencing analysis

A 411-bp fragment of the *rpoB* gene, containing the sequence of the 81-bp segment of *rpoB* gene, was amplified by PCR using primers *rpoB*-F (5'-TACGGTCGGC-GAGCTGATCC-3'). PCR was carried out in 50 μ L containing 2 μ L KCl, 2 μ L Tris pH 8.0, 1.5 μ L MgCl₂, 5 μ L dNTP, 1 U Taq polymerase, 27 μ L water (for molecular method), 20 pmol of each primer and 6–10 μ L DNA template. For sequencing the same primers were used with PCR parameters; 33 cycles of denaturation at 94°C for 30 s; primer annealing at 48°C for 45 s; extension at 60°C for 4 min. The *rpoB* gene fragments of *M. tuberculosis* iso-

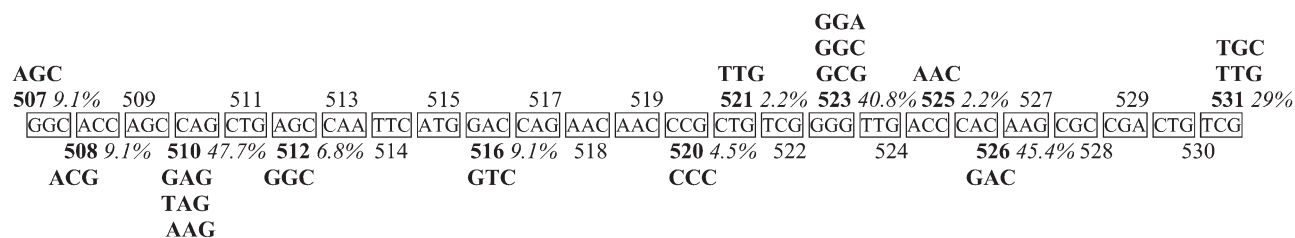


Figure 1. Percentage of mutation in different codons of the *rpoB* gene from 44 rifampicine-resistant strains of *M. Tuberculosis* isolated in Belarus

lates were sequenced using the Amersham autosequencer and Amersham Pharmacia DYEnamic ET Terminator Cycle Sequencing Premix kits. Alignment of the DNA fragments (rpoB) was carried out with the help of MEGA software (Blast program, Gen bank_PUBMED).

3 Results

All samples were cultured and identified as *M. tuberculosis* by biochemical and PCR method. All 44 isolates examined were resistant to isoniazid, rifampicine, streptomycin (100%) and 18 isolates (48%) were resistant to ethambutol. Mono-resistance to rifampicine was not observed in this study.

The hypervariable region (hot-spot) of *rpoB* gene was subjected to DNA sequencing analysis for the 44 rifampicine-resistant of *M. tuberculosis* clinical isolates (including MDR isolates). Sixteen different types of mutations were identified (Table 1). Fourteen silent mutations were found in four *M. tuberculosis* isolates in codons 520 (CCG>CCC) ($n=2$), 521 (CTG>TTG) ($n=1$), 508 (ACC>ACG) ($n=2$), 523 (GGG>GGA) ($n=1$), (GGG>GGC) ($n=4$) and 510 (CAG>TAG) ($n=4$) contributing to 16.8% of all mutations. All codons studied revealed point mutation, with the exception of codon 531, which indicate one substitution mutation (TCG>TGC). Point mutations were most frequently found in codons 510, 526 and 523 with frequencies of 47.7%, 40.9%, and 38.6%, respectively (Fig. 1). Eleven isolates (27.7%) carried one mutation and 23 (52%), 7 (16%), 4 (7%) carried two, three or four mutations, respectively (Fig. 1, Table 1).

4 Discussion

The mutations observed in this study is partially comparable to those reported from other countries (Ukraine, Russia, Italy, Turkey, USA, Poland) [15–17]. The codons 531, 526, 516 and 511 of *rpoB* gene are reported as the most frequently mutated sites worldwide, although variations in the relative frequencies of mutations in these codons have been described for isolates from different geographical locations [7, 11, 18]. CAG mutation (deletion, CTG, CAC or CAT) in codon 510 is very seldom identified in other countries; in contrast to other countries, our study indicates (Table 1) many more mutations of this kind (CAG→GAG, TAG, or AAG) in codon 510 [19, 20]. In Poland the mutation CAG→CAC has been found [7], in Russia CAG→CAT, but in Lithuania no mutations have occurred in this codon [21, 22].

It should be mentioned that mutations in codons 531 and 526 are the most frequently reported mutations worldwide (TCG→TTG for codon 531, and CAC→TAC for 526) [14]. There are other changes found in codons 531 (in Poland TCG→TGG, in Russia TCG→TGG, CAG or TGT,

in Lithuania TCG→TTG or TGG, in Italy TCG→TGG or TTG) and 526 (in Poland CAC→AAC, GAC or TGC, in Russia CAC→CTC, GAC, CAA, CAG, TGC, AAC, CGC or CCC, in Lithuania CAC→GAC, in Italy CAC→AAC, CTC, TAC or GAC). However, in our study the change TTG in codon 531 were found in 92%, and the change TGC in 8% of all isolates. In this study we have not detected any change of CAC→TAC in codon 526, but the change GAC 70% and CTC 30% was found in the isolates [21–23]. Our results correspond in part with those from previous reports [9, 24], but not those of Bárfai *et al.* [7]. 13.4% of our isolates carried a Ser-531-Leu substitution as one of the frequent mutations detected [3, 21, 22, 25].

While mutation of CAC→GAC at codon 526 occurred in 40.1% in Italian isolates [23] and 17.6% in Greece isolates [26], the mutation of CAC→TAC at codon 526 was dominant in American isolates (27.9%) [24] and Brazilian isolates (11%) [15]. Our data indicate very similar frequencies of His-526-Asp (45.5%) to Italian observations (40.1%) [23, 26–28].

Comparison of the data reported here with those in [29] indicate that codon 516 (GAC→GTC) occurs in lower frequencies (9.1%) and have different higher relative frequencies in isolates from different countries [7, 15–17]. In addition, Bárfai *et al.* [7], who analyzed Hungarian isolates, and Hirano *et al.* [29], who analyzed isolates mostly from Asian countries, found higher frequencies of mutation of GAC→GTC at codon 516 compared to our study. We could not detect any mutations in codon 511, which is one of most frequent mutations reported worldwide. These findings demonstrate that the frequencies of particular mutations in rifampicine-resistant *M. tuberculosis* isolates from Belarus are different from those that have been reported for isolates from other parts of the world.

Although the combination of two single point mutations has been described previously for rifampicine-resistant isolates of *M. tuberculosis* [1, 29]. The high percentage of double mutations found among the isolates in Belarus differed clearly from the lower prevalence of double mutations in other studies. A prominent finding of this study was the high frequency of double (52%) and triple (16%) and quadruple (7%) mutations occurring in separate codons (20%) (Table 1). We detected 15 (16.8%) nonsense mutations in 8 different isolates that demonstrated no effect in drug resistant pattern to rifampicine. All rifampicine-resistant isolates studied had mutations of different types in the *rpoB* region; however, we do not know which combination of specific types of mutations are associated with rifampicine resistance. Among the quadruple mutations detected in this study (3 isolates), one isolate showed a silent mutation in codon 508, and one had three silent mutation in codons 523, 521 and 520. Likewise, we observed among triple mutations (7 isolates) 2 isolates each with one silent mutation in codons 523 and 520; however, these silent mutations were demonstrated to have no effect in the drug-resistant pattern to ri-

Table 1. Frequency of amino acid and nucleotide changes of different codons in the *rpoB* gene of 44 rifampicine-resistant strains of *M. Tuberculosis* isolated in Belarus

Codon	Frequency		Amino acid change	Nucleotide change	Isolates
1 mutation					
526	4	2	His → Asp	CAC → GAC	74, 932
		2	His → Leu	CAC → CTC	384, 526
531	4	4	Ser→Leu	TCG→TTG	118, 367, 388,446
510	2	1	Gln→Glu	CAG→GAG	369
		1	Gln→Stop	CAG→TAG	23623
516	1	1	Asp→Val	GAC→GTC	407
2 mutations					
526	6	4	His→Asp	CAC→GAC	443, 507,489,2262
510			Gln→Glu	CAG→GAG	
		2	His→Leu	CAC→CTC	414,85
			Gln→Glu	CAG→GAG	
526	5	5	His→Asp	CAC → GAC	411, 440, 2548, 7285, 1416
523			Gly→Ala	GGG→GCG	
531		3	Ser→Leu	TCG→TTG	238, 447, 2331
510	4		Gln→Glu	CAG→GAG	
		1	Ser→Lys	TCG→TGC	1217
			Gln→Glu	CAG→GAG	
516		1	Asp→Val	GAC→GTC	2715
510	2		Gln→Stop	CAG→TAG	
		1	Asp→Val	GAC→GTC	370
			Gln→Glu	CAG→GAG	
523	1	1	Gly→Ala	GGG→GCG	139
516			Asp→Val	GAC→GTC	
512	1	1	Ser→ Gly	AGC→GGC	408
507			Arg→ Ala	GGC→GCC	
523	3	1	Ser→Leu	TCG→TTG	469
531			Gly →Ala	GGG→GCG	
		2	Ser→Leu	TCG→TTG	24276, 442
			Gly→Gly	GGG→GGC	
526	1	1	His→Asp	CAC→GAC	455
508			Thr→Thr	ACC→ACG	
3 mutations					
523	1	1	Gly→ Ala	GGC→ GCC	412
512			Ser→Gly	AGC→GGC	
508			Thr→Pro	ACC→CCC	
531	1	1	Ser→Leu	TCG→TTG	402
512			Ser→Gly	AGC→GGC	
507			Arg→Ala	GGC→GCC	
523	1	1	Gly→Gly	GGG→GCG	94
510			Gln→Lys	CAG→AAG	
507			Arg→Ser	GGC→AGC	
526	1	1	His→Asp	CAC→GAC	894
523			Gly→ Ala	GGG→GCG	
510			Gln→ Lys	CAG→AAG	
531	1	1	Ser→Leu	TCG→TTG	453
523			Gly→Ala	GGG→GCG	
526			His→Asp	CAC→GAC	
520	1	1	Pro→Pro	CCG→CCC	3255
510			Gln→Lys	CAG→AAG	
507			Gly→Ser	GGC→AGC	
525	1	1	Thr→Asn	ACC→AAC	3246
523			Gly→Ala	GGG→GCG	
510			Gln→ Stop	CAG→TAG	
4 mutation					
526	2	2	His→Asp	CAC→GAC	571, 471
523			Gly → Ala	GGG→GCG	
510			Gln →Stop,Glu	CAG→TAG,GAG	
508			Thr→Thr	ACC→ACG	
523	1	1	Gly→Gly	GGG→GGA	368
521			Leu →Leu	CTG→TTG	
520			Pro→Pro	CCG→CCC	
510			Gln→Glu	CAC→GAG	

fampicine (Table 1). Of the isolates, 33 (70.5%) were isolated from secondary cases (we do not have data on reactivation cases) (Table 1). Interestingly, all triple and quadruple mutations were found in isolates of sputum samples taken from patients with secondary infection. From 44 rifampicine-resistant isolates, 11 (29.5%) from sputum of patients with primary infection contained different types of mutations; in 5 isolates, 4 found to have single mutation in codon 531 TCG→TTG and 1 in codon 510 CAG→GAG. Among the cases of primary infection, 5 isolates indicated double mutations, in which 3 are in codons 531 TCG→TTG and 510 CAG→GAG. Of the 2 remaining isolates, 1 had a double mutation in codons 531 TCG→TTG and 523 GGC→GCG, and the other in codons 512 AGC→GCC and 507 GGC→GCC. The last isolate was found to have a triple mutation in codons 520 (silent) CCG→CCC, 507 GGC→AGC and 510 CAG→AAG. Eight patients with primary infection (72%) were shown to have predominant mutations in codon 531 TCG→TTG, and 5 (45.4%) in codon 510, and in patients with secondary infection 20 (60.6%) were found to have mutations in codon 526, 16 (48.4%) in codon 523 and 13 (39.3%) in codon 510 (Table 1). Thus, in Belarus, the highest frequency of common mutation shearing between primary and secondary infections in this study was found to occur in codon 510.

In bacterial populations, the generation of antibiotic resistance depends on the rate of emergence of resistant mutants [1]. A correlation between high mutation rate, antibiotic resistance and virulence in bacteria has been reported in several studies [4, 5]. Isolates of *M. tuberculosis* isolated in Belarus have been shown to display a wide spectrum of important mutations and might belong to the epidemic widespread clones. We will continue our investigation on the frequency of mutation in other genes responsible for resistance of *M. tuberculosis* to drugs and their association with the virulence of bacteria.

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